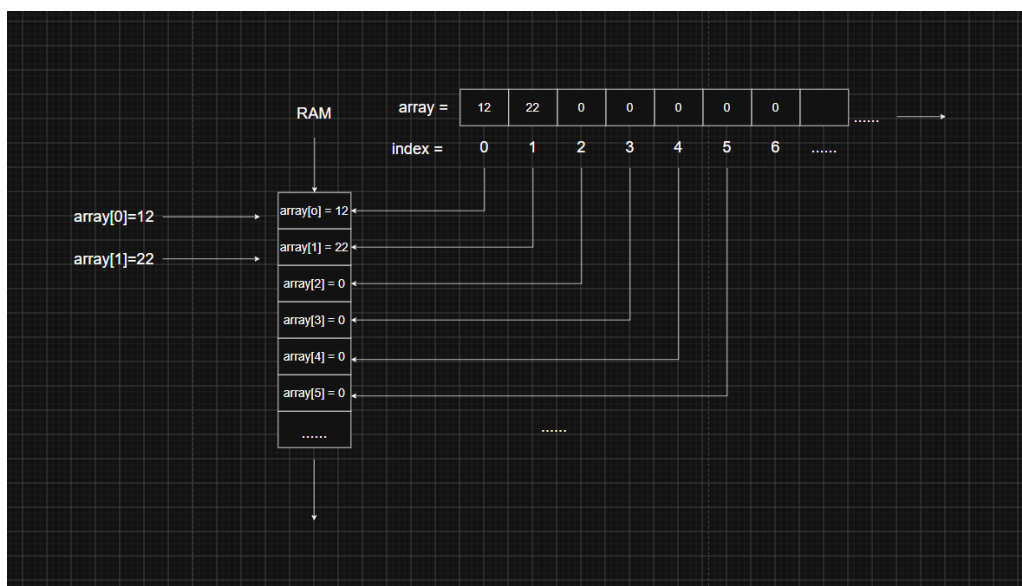
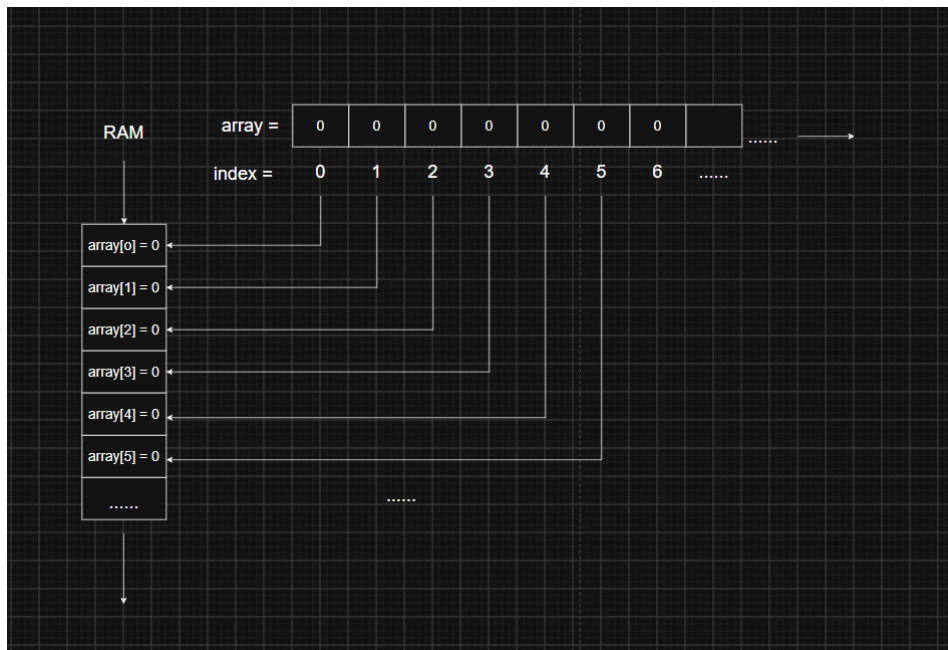


Q1.

1,4----

```
Example1.java  Example2.java  Example3.java  Example4.java  Example5.java  Ex
1  class Example{
2      public static void main(String args[]){
3          int [] arr = {12,35,47,59,62,73,81,93,111,121};
4
5          System.out.println(arr[0]);
6          System.out.println(arr[3]);
7          System.out.println(arr[5]);
8      }
9  }
10
```



Q2.

```
mple1.java Example2.java Example3.java Example4.java Example5.java Example6.java Example7.java
class Example2{
    public static void main(String args[]){
        double [] arr = {85.66, 78, 67.33, 57.66, 81, 91, 53.66, 61.66, 74.33, 79.33};
        System.out.println("average marks of the 3rd student "+arr[2]);
        System.out.println("average marks of the 7th student "+arr[6]);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
average marks of the 3rd student 67.33
average marks of the 7th student 53.66

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q3.

```
ample1.java ✕ Example2.java ✕ Example3.java ✕ Example4.java ✕ Example5.java ✕ Example6.java ✕
import java.util.*;
class Example3{
    public static void main(String args[]){
        int[] fX ={2362,2311,1917,1912,1887,1854,1717,1717,1779,1705,1806,1896};
        System.out.printf("Month\t\tUSD Million\n");
        System.out.printf("January\t\t%14s\n",fX[0]);
        System.out.printf("February\t\t%14s\n",fX[1]);
        System.out.printf("March\t\t%14s\n",fX[2]);
        System.out.printf("April\t\t%14s\n",fX[3]);
        System.out.printf("May\t\t%14s\n",fX[4]);
        System.out.printf("June\t\t%14s\n",fX[5]);
        System.out.printf("July\t\t%14s\n",fX[6]);
        System.out.printf("August\t\t%14s\n",fX[7]);
        System.out.printf("September\t\t%14s\n",fX[8]);
        System.out.printf("October\t\t%14s\n",fX[9]);
        System.out.printf("November\t\t%14s\n",fX[10]);
        System.out.printf("December\t\t%14s\n",fX[11]);
        System.out.println();

        fX [3]=1812;
        fX [6]=1817;

        System.out.println("April-"+fX[3]);
        System.out.println("July-"+fX[6]);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  + v
Month          USD Million
January        2362
February       2311
March          1917
April          1912
May            1887
June           1854
July           1717
August         1717
September      1779
October        1705
November       1806
December       1896

April-1812
July-1817

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q4.

```
import java.util.*;
class Example4{
    public static void main(String args[]){
        char [] ar ={'I','n','s','t','i','t','u','t','e',
                    'o','f','c','o','m','p','u','t','e','r','E','n',
                    'g','i','n','g','e','n','i','n','g','T','e','c','h','n','o','l','o','g','y'};

        System.out.println(""+ar[0]+ar[1]+ar[2]+ar[3]+ar[4]+ar[5]+ar[6]+ar[7]+ar[8]);
        System.out.println(""+ar[9]+ar[10]);
        System.out.println(""+ar[11]+ar[12]+ar[13]+ar[14]+ar[15]+ar[16]+ar[17]+ar[18]);
        System.out.println(""+ar[19]+ar[20]+ar[21]+ar[22]+ar[23]+ar[24]+ar[25]+ar[26]+ar[27]+ar[28]+ar[29]);
        System.out.println(""+ar[30]+ar[31]+ar[32]+ar[33]+ar[34]+ar[35]+ar[36]+ar[37]+ar[38]+ar[39]);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Institute
Of
Computer
Engineering
Technology

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q5.

```
Example1.java ✕ Example2.java ✕ Example3.java ✕ Example4.java ✕ Example5.java ✕ Example6.java ✕ Example7.java
1  import java.util.*;
2  class Example5{
3      public static void main(String args[]){
4          int [] marks = {56, 81, 43, 69, 93, 54, 48, 47, 51, 79, 82, 96, 57, 61, 66};
5          System.out.println(marks.length);
6          System.out.println("-----2");
7          for(int i=0;i<5;i++){
8              System.out.println(marks[i]);
9          }
10         System.out.println("-----3");
11         for(int i =0;i<marks.length;i++){
12             System.out.println(marks[i]);
13         }
14         System.out.println("-----4");
15         for(int i =9;i<marks.length;i++){
16             System.out.println(marks[i]);
17         }
18     }
19 }
20 }
21 }
22 }
```

Output.

```
15
-----2
56
81
43
69
93
-----3
56
81
43
69
93
54
48
47
51
79
82
96
57
61
66
-----4
79
82
96
57
61
66
```

Q6.

```
imple1.java Example2.java Example3.java Example4.java Example5.java Example6.java
import java.util.*;
class Example6{
    public static void main(String args[]){
        String [] day = new String[7];
        day[0]="Sunday";
        day[1]="Monday";
        day[2]="Tuesday";
        day[3]="Wednesday";
        day[4]="Thursday";
        day[5]="Friday";
        day[6]="Saturday";
        System.out.println(Arrays.toString(day));
        String [] new1 = new String[day.length];
        for(int i =0, y = day.length-1;i<day.length;i++,y--){
            new1[i]=day[y];
        }
        System.out.println(Arrays.toString(new1));
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
[Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday]
[Saturday, Friday, Thursday, Wednesday, Tuesday, Monday, Sunday]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q7.

```
ample1.java  Example2.java  Example3.java  Example4.java  Example5.java  Example6.java  Example7.java  Examp
import java.util.*;
class Example7{
    public static void main(String args[]){
        char [] bLetters = {'A','B','C','D','E','F','G','H','I','J',
                             'K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};
        System.out.println(Arrays.toString(bLetters));
        for(int i =0;i<bLetters.length;i+=2){
            System.out.print(bLetters[i]+" ");

        }
        System.out.println();

        for(int i =1;i<bLetters.length;i+=2){
            System.out.print(bLetters[i]+" ");
        }
        System.out.println();
        char [] new1 = new char[bLetters.length];
        for(int i =0,y = bLetters.length-1;i<bLetters.length;i++,y--){
            new1[i]=bLetters[y];
        }
        System.out.println(Arrays.toString(new1));
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
[A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z]
A C E G I K M O Q S U W Y
B D F H J L N P R T V X Z
[Z, Y, X, W, V, U, T, S, R, Q, P, O, N, M, L, K, J, I, H, G, F, E, D, C, B, A]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q8.

```
ample1.java  Example2.java  Example3.java  Example4.java  Example5.java  Example6.java  Example7.java  Example8.java
import java.util.*;
class Example8{
    public static void main(String args[]){
        int [] rainfallVal = {2346,1945,2060,1781,2365,1005,1162,1016,1512,2231,1903,12061,1005,1545,
                               2156,2037,1583,3668};

        System.out.println("the number of districts - "+rainfallVal.length);
        int y = 0;
        for(int i = 0; i < rainfallVal.length; i++){
            if(rainfallVal[i] >= 2000){
                y++;
            }
        }
        System.out.println("the number of districts with more than 2000mm - "+y);
        int z = 0;
        for(int i = 0; i < rainfallVal.length; i++){
            if(rainfallVal[i] < 1000){
                z++;
            }
        }
        System.out.println("the number of districts with less than 1000mm - "+z);
        double totRainVal = 0;
        for(int i = 0; i < rainfallVal.length; i++){
            totRainVal += rainfallVal[i];
        }
        double avg = (totRainVal/rainfallVal.length);
        System.out.println("the average annual rainfall values of all districts - "+avg);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
the number of districts - 18
the number of districts with more than 2000mm - 8
the number of districts with less than 1000mm - 0
the average annual rainfall values of all districts - 2410.0555555555557

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q9.

09. Which can be inserted at line 12, still the code will compile?

```
class Example{  
    public static void main(String args[]){  
        int[] array;  
        //Insert code here //Line 12  
    }  
}
```

A. array=new int[5];

C. array=new int[-5];

E. array=new int[] {10, 20, 30, 40, 50};

B. array=new int[10];

D. array={10, 20, 30, 40, 50};

F. array=new int[]{};

Answers – A,B,C,E,F.

Q10.

```
Example5.java  Example6.java  Example7.java  Example8.java  Example9.java  Example10.java  Example11.java
import java.util.*;
class Example10{
    public static void main(String args[]){
        int [] ar = {6,3,3,2,4,1,6,6,2,4};
        int z=0;
        int y=0;
        for(int i=0;i<ar.length;i++){
            if(ar[i]==3 | ar[i]==6){
                y++;
            }
        }
        for(int i=0;i<ar.length;i++){
            if(ar[i]%2==1){
                z++;
            }
        }
        System.out.println("the number of times '3' and '6' is rolled - "+y);
        System.out.println("the number of times odd numbers are rolled - "+z);
        boolean finder = true;
        for(int i=0;i<ar.length;i++){
            if(ar[i]==5){
                finder = true;
                break;
            }else{
                finder = false;
            }
        }
        if(finder==true){
            System.out.print("number 5 is rolled");
        }else{
            System.out.print("number 5 is not rolled");
        }
    }
}
```

Output.

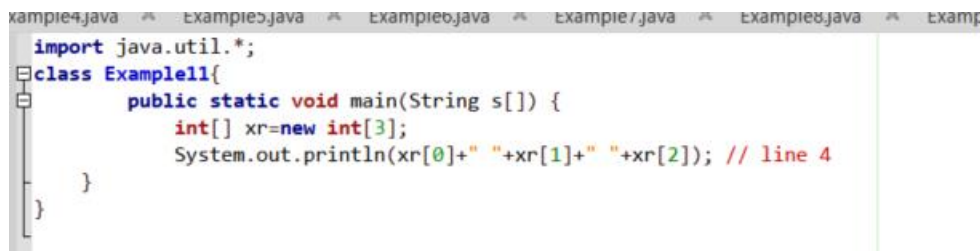
```
C:\WINDOWS\SYSTEM32\cmd  +  v
the number of times '3' and '6' is rolled - 5
the number of times odd numbers are rolled - 3
number 5 is not rolled

-----
(program exited with code: 0)

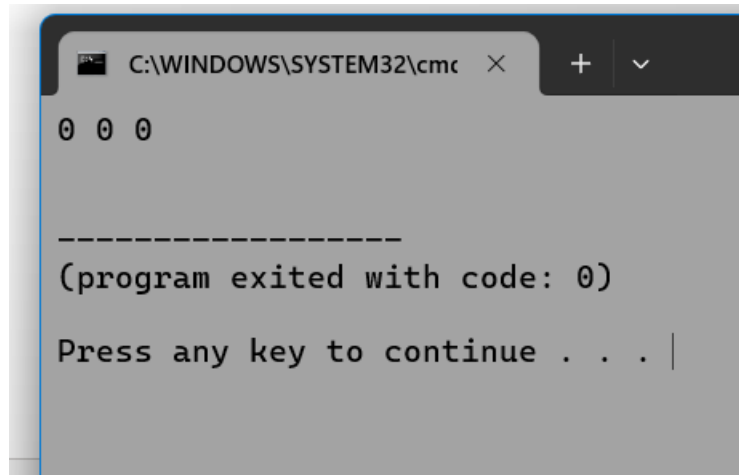
Press any key to continue . . . |
```

Q11.

1.



```
import java.util.*;
class Example11{
    public static void main(String s[]) {
        int[] xr=new int[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```



```
C:\WINDOWS\SYSTEM32\cmd
0 0 0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

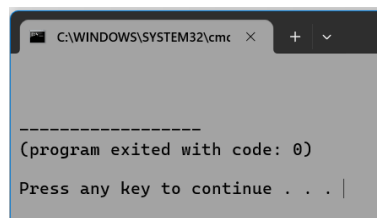
Explain – without assigning values, int arrays prints default value, it means 0.

2.

Char-



```
import java.util.*;
class Example11{
    public static void main(String s[]) {
        char[] xr=new char[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```



```
C:\WINDOWS\SYSTEM32\cmd
0 0 0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Boolean-

```
Example4.java Example5.java Example6.java Example7.java Example8.java
import java.util.*;
class Example11{
    public static void main(String s[]) {
        boolean[] xr=new boolean[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
false false false

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Double-

```
Example4.java Example5.java Example6.java Example7.java Example8.java
import java.util.*;
class Example11{
    public static void main(String s[]) {
        double[] xr=new double[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
0.0 0.0 0.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

String –

```
Example4.java Example5.java Example6.java Example7.java Example8.java
import java.util.*;
class Example11{
    public static void main(String s[]) {
        String[] xr=new String[3];
        System.out.println(xr[0]+" "+xr[1]+" "+xr[2]); // line 4
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
null null null

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q12.

```
Example5.java  Example6.java  Example7.java  Example8.java  Example9.java  Example10.java
import java.util.*;
class Example12{
    public static double tot=0;
    public static void main(String args[]){
        double [] ar = new double[9];
        Scanner sc = new Scanner(System.in);
        for(int i =0;i<ar.length;i++){
            System.out.print("Enter the Student "+(i+1)+" Marks : ");
            ar[i] =sc.nextDouble();
        }
        System.out.println();
        for(int i=0;i<ar.length;i++){
            tot += ar[i];
        }
        System.out.println("the total marks of Student - "+tot);
        System.out.println("the average mark of the student - "+tot/ar.length);
        double min = ar[0];
        for(int i=1;i<ar.length;i++){
            if(ar[i]<=min){
                min=ar[i];
            }
        }
        System.out.println("maximum mark of the student - "+min);
        double max = 0;
        for(int i=0;i<ar.length;i++){
            if(ar[i]>=max){
                max=ar[i];
            }
        }
        System.out.println("maximum mark of the student - "+max);
        int y = 0;
        for(int i=0;i<ar.length;i++){
            if(ar[i]>75){
                y++;
            }
        }
        System.out.println("how many marks are greater than 75 - "+y);
        int z = 0;
        for(int i=0;i<ar.length;i++){
            if(ar[i]<45){
                z++;
            }
        }
        System.out.println("how many marks are less than 45 - "+z);
        boolean finder = true;
        for(int i=0;i<ar.length;i++){
            if(ar[i]==100){
                finder = true;
                break;
            }else{
                finder = false;
            }
        }
        if(finder==true){
            System.out.print("Student get a 100 marks");
        }else{
            System.out.print("Student didn't get a 100 marks");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the Student 1 Marks : 23
Enter the Student 2 Marks : 4
Enter the Student 3 Marks : 56
Enter the Student 4 Marks : 76
Enter the Student 5 Marks : 87
Enter the Student 6 Marks : 100
Enter the Student 7 Marks : 45
Enter the Student 8 Marks : 7
Enter the Student 9 Marks : 78

the total marks of Student - 476.0
the average mark of the student - 52.888888888888886
maximum mark of the student - 4.0
maximum mark of the student - 100.0
how many marks are greater than 75 - 4
how many marks are less than 45 - 3
Student get a 100 marks

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter the Student 1 Marks : 23
Enter the Student 2 Marks : 4
Enter the Student 3 Marks : 45
Enter the Student 4 Marks : 45
Enter the Student 5 Marks : 23
Enter the Student 6 Marks : 12
Enter the Student 7 Marks : 34
Enter the Student 8 Marks : 78
Enter the Student 9 Marks : 98

the total marks of Student - 362.0
the average mark of the student - 40.222222222222222
maximum mark of the student - 4.0
maximum mark of the student - 98.0
how many marks are greater than 75 - 2
how many marks are less than 45 - 5
Student didn't get a 100 marks

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q13.

13. ===== program A =====

```
import java.util.*;
class Demo{
    public static void main(String s[]) {
        Scanner input=new Scanner(System.in);
        int[] numbers=new int[10];
        for (int i = 0; i < 10; i++){
            System.out.print("Input number "+(i+1)+" : ");
            numbers[i]=input.nextInt();
        }
        //print numbers
        System.out.print("[");
        for (int i = 0; i < 10; i++){
            System.out.print(numbers[i]+" , ");
        }
        System.out.println("\b\b]");
    }
}
```

===== program B =====

```
import java.util.*;
class Demo{
    public static void main(String s[]) {
        Random r=new Random ();
        int[] numbers=new int[10];
        for (int i = 0; i < 10; i++){
            numbers[i]=r.nextInt(); // Line X
        }
        //print numbers
        System.out.print("[");
        for (int i = 0; i < 10; i++){
            System.out.print(numbers[i]+" , ");
        }
        System.out.println("\b\b]");
    }
}
```

- I. What is the difference between A and B programs?
- II. Write how Line X changes to get numbers between 0 and 100.

Answers

1.A program has a scanner to get user inputs.

B program has random to generate random numbers.

2.

numbers[i]=r.nextInt(101);

Q14.

```
import java.util.*;
class Example14{
    public static void main(String args[]){
        Random r = new Random();
        int [] ar = new int[20];
        for(int i =0;i<ar.length;i++){
            ar[i] = r.nextInt(101);
        }
        int y = 0;
        System.out.println(Arrays.toString(ar));
        for(int i = 0;i<ar.length;i++){
            if(ar[i]>=50){
                y++;
            }
        }
        System.out.println("how many students pass the assignment - "+y);
        boolean finder = true;
        for(int i=0;i<ar.length;i++){
            if(ar[i]>=90){
                finder = true;
                break;
            }else{
                finder = false;
            }
        }
        if(finder==true){
            System.out.print("there are students with more than 90 marks");
        }else{
            System.out.print("there are no students with more than 90 marks");
        }
    }
}
```

Output.

```
[98, 76, 57, 2, 80, 81, 6, 77, 80, 30, 18, 83, 11, 58, 100, 19, 96, 44, 97, 15]
how many students pass the assignment - 12
there are students with more than 90 marks

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q15.

```
import java.util.*;
class Example15{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        Random r = new Random();
        System.out.print("Enter the array length : ");
        int z = sc.nextInt();

        int [] ar = new int[z];
        for(int i =0;i<ar.length;i++){
            ar[i] = r.nextInt(101);
        }
        int y = 0;
        System.out.println(Arrays.toString(ar));

        for(int i = 0;i<ar.length;i++){
            if(i%2==1){
                if(ar[i]%2==1){
                    y++;
                }
            }
        }
        System.out.println("how many odd numbers are in the odd index of an array - "+y);
        boolean finder = true;
        for(int i=0;i<ar.length;i++){
            if(ar[i]==100){
                finder = true;
                break;
            }else{
                finder = false;
            }
        }
        if(finder==true){
            System.out.print("number 100 is there");
        }else{
            System.out.print("number 100 is not there");
        }
        System.out.println();
        double min = ar[0];
        for(int i=1;i<ar.length;i++){
            if(ar[i]<=min){
                min=ar[i];
            }
        }
        System.out.println();
        System.out.println("maximum mark of the student - "+min);
        double max = 0;
        for(int i=0;i<ar.length;i++){
            if(ar[i]>=max){
                max=ar[i];
            }
        }
        System.out.println();
        System.out.println("maximum mark of the student - "+max);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter the array length : 6
[99, 59, 90, 94, 79, 85]
how many odd numbers are in the odd index of an array - 2
number 100 is not there

maximum mark of the student - 59.0

maximum mark of the student - 99.0
```

Q16.

1.

```
Example11.java Example12.java Example13.java Example14.java Example15.java
import java.util.*;
class Example16{
    public static void main(String args[]){
        int[] num={10,20,30,40,50,60,70};
        System.out.println("Length : "+num.length);
        for(int i=0; i<=num.length; i++){
            System.out.println("Element at index " + i + " is " + num[i]);
        }
    }
}
```

Output 1

```
Length : 7
Element at index 0 is 10
Element at index 1 is 20
Element at index 2 is 30
Element at index 3 is 40
Element at index 4 is 50
Element at index 5 is 60
Element at index 6 is 70
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 7 out of bounds for length 7
    at Example16.main(Example16.java:7)
```

Explain – there is no index 7 in array

2.

```
Example11.java Example12.java Example13.java Example14.java Example15.java
1 import java.util.*;
2 class Example16{
3     public static void main(String args[]){
4         int[] num={10,20,30,40,50,60,70};
5         System.out.println("Length : "+num.length);
6         for(int i=0; i<=num.length-1; i++){
7             System.out.println("Element at index " + i + " is " + num[i]);
8         }
9     }
10 }
11 }
12 }
13 }
```

Output

```
C:\WINDOWS\SYSTEM32\cmd.exe
Length : 7
Element at index 0 is 10
Element at index 1 is 20
Element at index 2 is 30
Element at index 3 is 40
Element at index 4 is 50
Element at index 5 is 60
Element at index 6 is 70

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q17.

```
Example12.java Example13.java Example14.java Example15.java Example16.java E
1 import java.util.*;
2 class Example17{
3     public static void main(String args[]){
4         int[] arr={10,20,30,40,50,60,70};
5         int[] num=arr;
6         System.out.println(Arrays.toString(num)); //Line 1
7         System.out.println(Arrays.toString(arr)); //Line 2
8         System.out.println("num == arr ??? "+ (num==arr)); //Line 3
9         System.out.println(arr); //Line 4
10    }
11 }
12 }
```

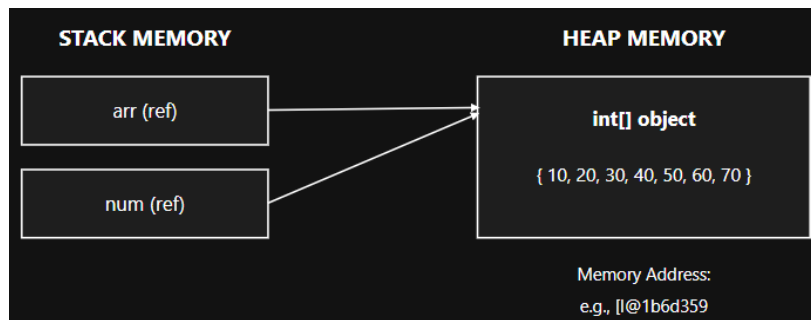
```
C:\WINDOWS\SYSTEM32\cmd X + v
[10, 20, 30, 40, 50, 60, 70]
[10, 20, 30, 40, 50, 60, 70]
num == arr ??? true
[I@6842775d

-----
(program exited with code: 0)
Press any key to continue . . .
```

Line 3 - num and arr point to the exact same array in memory, so num == arr evaluates to true

Line 4 - Printing an array variable directly calls Object.toString(), which outputs the type ([I for int[]) followed by @ and its hexadecimal identity hash code.

```
import java.util.*;
class Example17{
    public static void main(String args[]){
        int[] arr={10,20,30,40,50,60,70};
        int[] num=arr;
        System.out.println(Arrays.toString(num)); //Line 1
        System.out.println(Arrays.toString(arr)); //Line 2
        System.out.println("num == arr ??? "+ (num==arr)); //Line 3
        System.out.println(arr); //Line 4
        System.out.println(num); //Line 5
    }
}
```



Q18.

```
Example13.java Example14.java Example15.java Example16.java Example17.java Example18.java
1 import java.util.*;
2 class Example18{
3     public static void main(String args[]){
4         String[] month = {"January", "February", "March", "April", "May", "June",
5             "July", "August", "September", "October", "November", "December"};
6         String[] arrMonth = new String[month.length];
7         for (int i = 0; i < month.length; i++) {
8             arrMonth[i] = month[i];
9         }
10        System.out.println("month == arrMonth ??? " + (month==arrMonth));
11    }
12 }
13 }
```

```
C:\WINDOWS\SYSTEM32\cmd
month == arrMonth ??? false

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Explain –

== compares memory addresses. Since arrMonth is created as a **new** array in heap memory, it has a different memory location than month, so it returns false.

Q19.

```
import java.util.*;
class Example19{
    public static void main(String args[]){
        Random r = new Random();
        int [] ar = new int[50];
        for(int i =0;i<ar.length;i++){
            ar[i]=r.nextInt();
        }
        System.out.println(Arrays.toString(ar));
        System.out.println();
        int [] new1 = new int [ar.length];
        for(int i =0,y=ar.length-1;i<ar.length;i++,y--){
            new1 [i] = ar[y];
        }
        System.out.println(Arrays.toString(new1));
        int negativeCount = 0;
        for (int num : ar) {
            if (num < 0) {
                negativeCount++;
            }
        }
        System.out.println();
        double probability = (double) negativeCount / ar.length;
        System.out.println("Probability of negative numbers: " + probability);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
[838857585, -1285650120, -1630323931, 1680127382, -1251411374, -1698420625, 898993334, -410733492, -2070615682, -256427319, 500540347, 17572
85482, -1529040156, 363235911, 1501798658, 768154127, 27482234, 437814329, -1477942888, -1721152167, 56852541, 773773041, 1400855658, 423747
900, -298597258, -1233537350, 182335022, -1652642924, -1442233283, 276168200, -1624821885, 1068587644, -176932798, -1232036624, -894316202,
129282784, 1829408715, 1771899866, 392657800, -529750960, 1919998456, -1814841996, 1141357005, 341001751, -2043543730, -670528535, -32908829
7, -1339027060, -1952805006, -1499906734]

[-1499906734, -1952805006, -1339027060, -329088297, -670528535, -2043543730, 341001751, 1141357005, -1814841996, 1919998456, -529750960, 392
657800, 1771899866, 1829408715, 129282784, -894316202, -1232036624, -176932798, 1068587644, -1624821885, 276168200, -1442233283, -1652642924
, 182335022, -1233537350, -298597258, 423747900, 1400855658, 773773041, 56852541, -1721152167, -1477942888, 437814329, 27482234, 768154127,
1501798658, 363235911, -1529040156, 1757285482, 500540347, -256427319, -2070615682, -410733492, 898993334, -1698420625, -1251411374, 1680127
382, -1630323931, -1285650120, 838857585]

Probability of negative numbers: 0.52

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q20.

```
Example17.java Example18.java Example19.java Example20.java Example21.java Exam
import java.util.*;
class Example20{
    public static void main(String args[]){
        char[] vowels1 = {'A', 'E', 'I', 'O', 'U'};
        char[] vowels2 = {'a', 'e', 'i', 'o', 'u'};
        char[] ar1 = new char[vowels1.length + vowels2.length];
        for (int i = 0; i < vowels1.length; i++) {
            ar1[i] = vowels1[i];
            ar1[i + vowels1.length] = vowels2[i];
        }
        for (char c : ar1)
            System.out.print(c + " ");
        System.out.println();

        char [] ar2 = new char[vowels1.length+vowels2.length];
        for (int i = 0; i < vowels1.length; i++) {
            ar2[2 * i] = vowels1[i];
            ar2[2 * i + 1] = vowels2[i];
        }
        for (char c : ar2)
            System.out.print(c + " ");
        System.out.println();
        char[] ar3 = new char[vowels1.length + vowels2.length];
        for (int i = 0; i < vowels1.length; i++) {
            ar3[2 * i] = vowels1[vowels1.length - 1 - i];
            ar3[2 * i + 1] = vowels2[vowels2.length - 1 - i];
        }
        for (char c : ar3) |
            System.out.print(c + " ");
        System.out.println();
        char[] ar4 = new char[vowels1.length + vowels2.length];
        for (int i = 0; i < vowels1.length; i++) {
            ar4[2 * i] = vowels1[i];
            ar4[2 * i + 1] = vowels2[vowels2.length - 1 - i];
        }
        for (char c : ar4)
            System.out.print(c + " ");
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
A E I O U a e i o u
A a E e I i O o U u
U u O o I i E e A a
A u E o I i O e U a
-----
```

Q21.

```
import java.util.*;
class Example21{
    public static void main(String args[]){
        int[] ar1 = new int[10];
        double[] dr1 = new double[10];
        //new line
    }
}
```

Answers – C , E.

Q22.

22. What is the meaning of the phrase "wider conversion" in Java arrays?

- A) It refers to the process of converting a value from a narrower array type to a wider array type.
- B) It refers to the process of converting a value from a wider array type to a narrower array type.
- C) It refers to the process of converting an array from one reference type to another reference type.
- D) It refers to the process of converting an array from a primitive type to a reference type.



Answer – A.

Q23.

23. Which can be inserted at line 20, still the code will compile?

```
class Example{  
    public static void main(String args[]){  
        int x=0;  
        int[] xr=new int[3];  
        double d=0.0;  
        double[] dr=new double[5];  
        int[] grade={'a','b'};  
        //Insert code here //Line 20  
    }  
}
```

A. `x=xr[0];`

B. `xr[0]=x;`

C. `x=xr;`

D. `xr=x;`

E. `dr[0]=xr[0];`

F. `xr[0]=dr[0];`

G. `xr[0]=(int)dr[0];`

H. `xr=dr;`

I. `dr=(double[])xr;`

J. `dr=xr;`

K. `xr=(int)dr;`

L. `xr=(int[])dr;`

Answers – A,B,E,G.

Q24.

```
import java.util.*;
class Example24{
    public static void increment(int salary1, int salary2, int salary3, int salary4){
        salary1 +=5000;
        salary2 +=5000;
        salary3 +=5000;
        salary4 +=5000;
    }

    public static void main(String args[]){
        double[] salary={71,86,54,89};
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);//Line1
        increment(salary [0], salary [1], salary [2], salary [3]); //Line 2
        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
    }
}
```

```
javac "Example24.java" (in directory: D:\ICD124\ICET-\DAY 9\N)
Example24.java:13: error: incompatible types: possible lossy conversion from double to int
        increment(salary [0], salary [1], salary [2], salary [3]); //Line 2
                  ^
Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output
1 error
Compilation failed.
```

We cant run this without typecasting casting parameters

Correcting—

```
import java.util.*;
class Example24{
    public static void increment(double [] salary){
        for(int i =0;i<salary.length;i++){
            salary[i] +=5000;
        }
    }

    public static void main(String args[]){
        double[] salary={71,86,54,89};
        for (double s : salary) {
            System.out.print(s + " ");
        }
        System.out.println();
        increment(salary);
        for (double s : salary) {
            System.out.print(s + " ");
        }
    }
}
```

```
71.0 86.0 54.0 89.0
5071.0 5086.0 5054.0 5089.0
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

Q25.

```
1 import java.util.*;
2 class Example25{
3     public static void main(String args[]){
4         int [] dollar = {15000, 21000, 18500, 12275, 35550, 28760, 17175, 10500};
5         double [] rupee = new double [dollar.length];
6         for(int i =0;i<dollar.length;i++){
7             rupee[i] = ((double) dollar[i]) * 366.1;
8         }
9
10
11         int upvalue = 7500000;
12         int more_count = 0;
13         int less_count = 0;
14         for(int i =0;i<rupee.length;i++){
15             if(rupee[i]>=upvalue){
16                 more_count ++;
17             }else{
18                 less_count ++;
19             }
20         }
21         System.out.println("how many software engineers get more than Rs.7,500,000 : "+more_count);
22         System.out.println("how many software engineers get less than Rs.7,500,000 : "+less_count);
23
24         double [] uVal = new double[rupee.length];
25         double [] lVal = new double[rupee.length];
26         for(int i = 0,y=0;i<rupee.length;i++,y++){
27             if(rupee[i]>=upvalue){
28                 uVal[i]=rupee[i];
29             }else{
30                 lVal[i]=rupee[i];
31             }
32         }
33         System.out.println(Arrays.toString(uVal));
34         System.out.println(Arrays.toString(lVal));
35     }
36 }
37
38
```

5.java" (in directory: D:\ICD124\ICET-\DAY 9\N) 5
ished successfully.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
how many software engineers get more than Rs.7,500,000 : 3
how many software engineers get less than Rs.7,500,000 : 5
[0.0, 7688100.000000001, 0.0, 0.0, 1.3014855E7, 1.0529036E7, 0.0, 0.0]
[5491500.0, 0.0, 6772850.0, 4493877.5, 0.0, 0.0, 6287767.5, 3844050.0000000005]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q26.

```
import java.util.*;
class Example26{
    public static void sort(int[] ar) {
        for (int j = 0; j < 3; j++) {
            if (ar[j] > ar[j + 1]) {
                int t = ar[j];
                ar[j] = ar[j + 1];
                ar[j + 1] = t;
            }
        }

        for (int j = 0; j < 3; j++) {
            if (ar[j] > ar[j + 1]) {
                int t = ar[j];
                ar[j] = ar[j + 1];
                ar[j + 1] = t;
            }
        }

        for (int j = 0; j < 3; j++) {
            if (ar[j] > ar[j + 1]) {
                int t = ar[j];
                ar[j] = ar[j + 1];
                ar[j + 1] = t;
            }
        }
    }

    System.out.print(Arrays.toString(ar));
    for (int i = 0; i < ar.length - 1; i++) {
        for (int j = 0; j < ar.length - 1 - i; j++) {
            if (ar[j] > ar[j + 1]) {
                int t = ar[j];
                ar[j] = ar[j + 1];
                ar[j + 1] = t;
            }
        }
    }
}

public static void main(String[] args) {
    int[] arr = {96, 17, 54, 6, 81, 47, 63, 79, 32, 13};
    sort(arr);
    System.out.println();
    System.out.print(Arrays.toString(arr));
}
}
```

```
[6, 17, 54, 96, 81, 47, 63, 79, 32, 13]
[6, 13, 17, 32, 47, 54, 63, 79, 81, 96]
```

(program exited with code: 0)

Press any key to continue . . . |

Q27.

```
Example19.java  Example20.java  Example21.java  Example24.java  Example25.java  Example26.java  Example27.java
import java.util.*;
class Example27{
    public static void sortDollarSalary(int[] dollar) {
        for (int i = 0; i < dollar.length - 1; i++) {
            for (int j = 0; j < dollar.length - 1 - i; j++) {
                if (dollar[j] > dollar[j + 1]) {
                    int temp = dollar[j];
                    dollar[j] = dollar[j + 1];
                    dollar[j + 1] = temp;
                }
            }
        }
    }
    public static void main(String[] args) {
        int[] dollar = {15000, 21000, 18500, 12275, 35550, 28760, 17175, 10500}; //not23, it is 25 Question
        sortDollarSalary(dollar);
        System.out.println(Arrays.toString(dollar));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd  +  v
[10500, 12275, 15000, 17175, 18500, 21000, 28760, 35550]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q28.

```
Example20.java Example21.java Example24.java Example25.java Example26.java Example27.java
import java.util.*;
class Example28 {
    public static int[] modifyArray(int[] arr, int index, int value) {
        int[] newArr = new int[arr.length];
        for (int i = 0; i < arr.length; i++) {
            newArr[i] = arr[i];
        }
        if (index >= 0 && index < newArr.length) {
            newArr[index] = value;
        }
        return newArr;
    }
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] Array1 = {10, 20, 30, 40};
        System.out.println(Arrays.toString(Array1));
        System.out.print("Enter replace index: ");
        int index = input.nextInt();
        System.out.print("Enter replace value: ");
        int value = input.nextInt();
        int[] modifiedArray = modifyArray(Array1, index, value);
        System.out.println("Previous array -> " + Arrays.toString(Array1));
        System.out.println("New array -> " + Arrays.toString(modifiedArray));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
[10, 20, 30, 40]
Enter replace index: 2
Enter replace value: 55
Previous array -> [10, 20, 30, 40]
New array -> [10, 20, 55, 40]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q29.

```
Example21.java Example24.java Example25.java Example26.java Example27.java Example28.java Example29.java
import java.util.*;
class Q29 {
    public static String[] addSubject(String[] subject) {
        String[] temp = new String[subject.length + 1];

        for (int i = 0; i < temp.length; i++) {
            if (i < 7) {
                temp[i] = subject[i];
            } else if (i == 7) {
                temp[i] = "Music";
            } else {
                temp[i] = subject[i - 1];
            }
        }
        return temp;
    }

    public static void main(String[] args) {
        String[] subject = {"Mathematics", "Science", "English",
                            "Sinhala", "History", "Buddhisum", "Business Studies", "Computer Studies"};
        String[] result = addSubject(subject);
        System.out.println(Arrays.toString(result));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
[Mathematics, Science, English, Sinhala, History, Buddhisum, Business Studies, Music, Computer Studies]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q30.

```
import java.util.*;

class Example30 {
    public static void printData(String[] months, double[] reserves) {
        for (int i = 0; i < months.length; i++) {
            System.out.println(months[i] + " : " + reserves[i]);
        }
    }

    public static void sortDescending(String[] months, double[] reserves) {
        for (int i = 0; i < reserves.length - 1; i++) {
            for (int j = 0; j < reserves.length - 1 - i; j++) {
                if (reserves[j] < reserves[j + 1]) {
                    double tempReserve = reserves[j];
                    reserves[j] = reserves[j + 1];
                    reserves[j + 1] = tempReserve;
                    String tempMonth = months[j];
                    months[j] = months[j + 1];
                    months[j + 1] = tempMonth;
                }
            }
        }
    }

    public static void main(String[] args) {
        String[] originalMonths = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                                   "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
        double[] originalReserves = {2361.0, 2311.0, 1916.0, 1812.0, 1920.0,
                                     1859.0, 1818.0, 1716.0, 1776.0, 1703.0, 1805.0, 1896.0};

        String[] monthsCopy = new String[originalMonths.length];
        double[] reservesCopy = new double[originalReserves.length];

        for (int i = 0; i < originalMonths.length; i++) {
            monthsCopy[i] = originalMonths[i];
            reservesCopy[i] = originalReserves[i];
        }

        System.out.println("");
        printData(originalMonths, originalReserves);
        sortDescending(monthsCopy, reservesCopy);
        System.out.println("\n Reserves");
        printData(monthsCopy, reservesCopy);
    }
}
```

```
Jan : 2361.0
Feb : 2311.0
Mar : 1916.0
Apr : 1812.0
May : 1920.0
Jun : 1859.0
Jul : 1818.0
Aug : 1716.0
Sep : 1776.0
Oct : 1703.0
Nov : 1805.0
Dec : 1896.0
```

```
Reserves
Jan : 2361.0
Feb : 2311.0
May : 1920.0
Mar : 1916.0
Dec : 1896.0
Jun : 1859.0
Jul : 1818.0
Apr : 1812.0
Nov : 1805.0
Sep : 1776.0
Aug : 1716.0
Oct : 1703.0
```

Q31.

```
Example25.java Example26.java Example27.java Example28.java Example29.java Example30.java
import java.util.*;
class Example31 {
    public static void printList(String[] names, int[] marks) {
        for (int i = 0; i < names.length; i++) {
            System.out.println(names[i] + " - " + marks[i]);
        }
    }
    public static void main(String[] args) {
        String[] names = {"Sachini", "Tharusha", "Shehan", "Ashini", "Sachin", "Kasun",
            "Lihini", "Hashini", "Ashini", "Kaveesha"};
        int[] marks = {807, 796, 728, 801, 818, 746, 781, 756, 801, 812};
        printList(names, marks);
        String[] tempNames = new String[names.length + 1];
        int[] tempMarks = new int[marks.length + 1];
        for (int i = 0; i < names.length; i++) {
            tempNames[i] = names[i];
            tempMarks[i] = marks[i];
        }
        tempNames[names.length] = "Amesh";
        tempMarks[marks.length] = 804;

        System.out.println();
        printList(tempNames, tempMarks);
        int removeIndex = -1;
        for (int i = 0; i < tempNames.length; i++) {
            for (int j = i + 1; j < tempNames.length; j++) {
                if (tempNames[i].equals(tempNames[j])) {
                    removeIndex = j;
                    break;
                }
            }
        }
        String[] uniqueNames = new String[tempNames.length - 1];
        int[] uniqueMarks = new int[tempMarks.length - 1];
        for (int i = 0, k = 0; i < tempNames.length; i++) {
            if (i == removeIndex) continue;
            uniqueNames[k] = tempNames[i];
            uniqueMarks[k] = tempMarks[i];
            k++;
        }
        System.out.println();
        printList(uniqueNames, uniqueMarks);
        for (int i = 0; i < uniqueMarks.length - 1; i++) {
            for (int j = 0; j < uniqueMarks.length - 1 - i; j++) {
                if (uniqueMarks[j] < uniqueMarks[j + 1]) {
                    int tm = uniqueMarks[j];
                    uniqueMarks[j] = uniqueMarks[j + 1];
                    uniqueMarks[j + 1] = tm;

                    String tn = uniqueNames[j];
                    uniqueNames[j] = uniqueNames[j + 1];
                    uniqueNames[j + 1] = tn;
                }
            }
        }
        System.out.println();
        printList(uniqueNames, uniqueMarks);
    }
}
```

Sachini - 807
Tharusha - 796
Shehan - 728
Ashini - 801
Sachin - 818
Kasun - 746
Lihini - 781
Hashini - 756
Ashini - 801
Kaveesha - 812
Amesh - 804

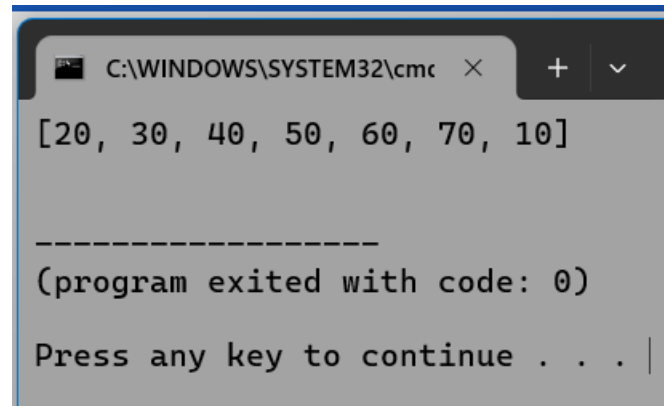
Sachini - 807
Tharusha - 796
Shehan - 728
Ashini - 801
Sachin - 818
Kasun - 746
Lihini - 781
Hashini - 756
Kaveesha - 812
Amesh - 804

Sachin - 818
Kaveesha - 812
Sachini - 807
Amesh - 804
Ashini - 801
Tharusha - 796
Lihini - 781
Hashini - 756
Kasun - 746
Shehan - 728

(program exited with code: 0)

Q32.

```
import java.util.*;
class Example32{
    public static void rotateLeft(int[] arr) {
        int first = arr[0];
        for (int i = 0; i < arr.length - 1; i++) {
            arr[i] = arr[i + 1];
        }
        arr[arr.length - 1] = first;
    }
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50, 60, 70};
        rotateLeft(arr);
        System.out.println(Arrays.toString(arr));
    }
}
```



```
C:\WINDOWS\SYSTEM32\cmd
[20, 30, 40, 50, 60, 70, 10]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q33.

```
Example27.java Example28.java Example29.java Example30.java
import java.util.*;
class Example33{
    public static void sortBinaryArray(int[] arr) {
        int zeroIndex = 0;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == 0) {
                int temp = arr[i];
                arr[i] = arr[zeroIndex];
                arr[zeroIndex] = temp;
                zeroIndex++;
            }
        }
    }
    public static void main(String[] args) {
        int[] arr = {1, 0, 1, 1, 0, 0, 1, 0};
        sortBinaryArray(arr);
        System.out.println(Arrays.toString(arr));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
[0, 0, 0, 0, 1, 1, 1, 1]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q34.

```
import java.util.*;

class Example34 {
    public static void printList(String[] countries, double[] times) {
        for (int i = 0; i < countries.length; i++) {
            System.out.println(countries[i] + " - " + times[i] + "s");
        }
    }

    public static void main(String[] args) {
        String[] countries = {
            "Switzerland(2)", "Ivory Coast(1)", "Jamaica(2)", "United States(1)",
            "United States(2)", "Great Britain", "Ivory Coast(2)", "Jamaica(3)",
            "Switzerland(1)", "Jamaica(1)"
        };
        double[] times = {11.11, 10.86, 10.76, 10.98, 11.11, 11.05, 11.14, 10.61, 11.06, 10.74};
        int minIndex = 0;
        for (int i = 1; i < times.length; i++) {
            if (times[i] < times[minIndex]) {
                minIndex = i;
            }
        }
        System.out.println("Gold Medal Winner 2021: " + countries[minIndex] + " - " + times[minIndex] + "s");
        int secondIndex = -1;
        for (int i = 0; i < times.length; i++) {
            if (i == minIndex) continue;
            if (secondIndex == -1 || times[i] < times[secondIndex]) {
                secondIndex = i;
            }
        }
        System.out.println("\nNew Gold Medal Winner 2022: " + countries[secondIndex] + " - " + times[secondIndex] + "s");
        String[] newCountries = new String[countries.length - 1];
        double[] newTimes = new double[times.length - 1];

        for (int i = 0, k = 0; i < countries.length; i++) {
            if (i == minIndex) continue;
            newCountries[k] = countries[i];
            newTimes[k] = times[i];
            k++;
        }

        for (int i = 0; i < newTimes.length - 1; i++) {
            for (int j = 0; j < newTimes.length - 1 - i; j++) {
                if (newTimes[j] > newTimes[j + 1]) {
                    double tempTime = newTimes[j];
                    newTimes[j] = newTimes[j + 1];
                    newTimes[j + 1] = tempTime;

                    String tempCountry = newCountries[j];
                    newCountries[j] = newCountries[j + 1];
                    newCountries[j + 1] = tempCountry;
                }
            }
        }

        System.out.println();
        printList(newCountries, newTimes);
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd  + v

Gold Medal Winner 2021: Jamaica(3) - 10.61s

New Gold Medal Winner 2022: Jamaica(1) - 10.74s

Jamaica(1) - 10.74s
Jamaica(2) - 10.76s
Ivory Coast(1) - 10.86s
United States(1) - 10.98s
Great Britain - 11.05s
Switzerland(1) - 11.06s
Switzerland(2) - 11.11s
United States(2) - 11.11s
Ivory Coast(2) - 11.14s

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q35.

```
import java.util.*;
class Example35{
    public static void findDuplicate(int[] arr) {
        boolean dup = false;
        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    System.out.println("duplicate \"" + arr[i] + "\"");
                    dup = true;
                    return;
                }
            }
        }
        if (!dup) System.out.println("no duplicate");
    }

    public static boolean equals(int[] a, int[] b) {
        if (a.length != b.length) return false;
        for (int i = 0; i < a.length; i++) {
            if (a[i] != b[i]) return false;
        }
        return true;
    }

    public static void main(String[] args) {
        int[] ar={10,20,30,40,50};
        int[] br={10,20,10,40,50,60};
        int[] cr={30,30,30,30};
        int[] dr={40,10,50,20,30};
        int[] er={10,20,30,40,50};
        int[] fr={10,20,30,40};

        findDuplicate(ar);
        findDuplicate(br);
        findDuplicate(cr);
        findDuplicate(dr);
        findDuplicate(er);
        findDuplicate(fr);

        System.out.println("ar==br : " + equals(ar,br));
        System.out.println("ar==cr : " + equals(ar,cr));
        System.out.println("ar==dr : " + equals(ar,dr));
        System.out.println("ar==er : " + equals(ar,er));
        System.out.println("ar==fr : " + equals(ar,fr));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
no duplicate
duplicate "10"
duplicate "30"
no duplicate
no duplicate
no duplicate
ar==br : false
ar==cr : false
ar==dr : false
ar==er : true
ar==fr : false

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q36.

```
import java.util.*;
class Example36 {
    public static void main(String[] args) {
        String[] cricket = {"Shanaka", "Dinesh", "Amesh", "Lahiru", "Dilshan",
                           "Hashan", "Wenura", "Randika", "Dasun", "Thisara", "Kasun", "Nisal", "Praveen"};
        String[] football = {"Anjana", "Chamara", "Kamal", "Dhanuka", "Shanaka",
                            "Kasun", "Dineth", "Sahan", "Wenura", "Amesh", "Mahesh", "Randika", "Thisara"};
        String[] both = new String[cricket.length];
        int countBoth = 0;

        for (int i = 0; i < cricket.length; i++) {
            for (int j = 0; j < football.length; j++) {
                if (cricket[i].equals(football[j])) {
                    both[countBoth++] = cricket[i];
                    break;
                }
            }
        }
        String[] resultBoth = new String[countBoth];
        for (int i = 0; i < countBoth; i++) {
            resultBoth[i] = both[i];
        }
        System.out.println("Both Cricket and Football: " + Arrays.toString(resultBoth));
        String[] either = new String[cricket.length + football.length];
        int countEither = 0;
        for (int i = 0; i < cricket.length; i++) {
            either[countEither++] = cricket[i];
        }
        for (int i = 0; i < football.length; i++) {
            boolean exists = false;
            for (int j = 0; j < countEither; j++) {
                if (football[i].equals(either[j])) {
                    exists = true;
                    break;
                }
            }
            if (!exists) {
                either[countEither++] = football[i];
            }
        }
        String[] resultEither = new String[countEither];
        for (int i = 0; i < countEither; i++) {
            resultEither[i] = either[i];
        }
        System.out.println("Cricket or Football: " + Arrays.toString(resultEither));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
Both Cricket and Football: [Shanaka, Amesh, Wenura, Randika, Thisara, Kasun]
Cricket or Football: [Shanaka, Dinesh, Amesh, Lahiru, Dilshan, Hashan, Wenura, Randika, Dasun, Thisara, Kasun, Nisal, Praveen, Anjana, Chamara, Kamal, Dhanuka, Dineth, Sahan, Mahesh]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q37.

```
import java.util.*;
class Example37 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String text = input.nextLine();
        char[] charArray = new char[text.length()];
        for (int i = 0; i < text.length(); i++) {
            charArray[i] = text.charAt(i);
        }
        System.out.println("Char Array: " + Arrays.toString(charArray));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a string: sdfdsdf
Char Array: [s, d, f, s, d, s, f]

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q38.

```
Example31.java Example32.java Example33.java Example34.java Example35.java Example36.java
import java.util.*;
class Example38 {
    public static boolean hasSameDigits(int n1, int n2) {
        char[] a1 = Integer.toString(n1).toCharArray();
        char[] a2 = Integer.toString(n2).toCharArray();
        Arrays.sort(a1);
        Arrays.sort(a2);
        return Arrays.equals(a1, a2);
    }
    public static void main(String[] args) {
        int x = 1;
        while (true) {
            if (hasSameDigits(x, 2 * x) &&
                hasSameDigits(x, 3 * x) &&
                hasSameDigits(x, 4 * x) &&
                hasSameDigits(x, 5 * x) &&
                hasSameDigits(x, 6 * x)) {
                System.out.println("Smallest positive integer x: " + x);
                break;
            }
            x++;
        }
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
Smallest positive integer x: 142857
```

```
-----
(program exited with code: 0)
```

```
Press any key to continue . . . |
```

Q39.

```
import java.util.*;
class Example39 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] ids = new int[100];
        String[] names = new String[100];
        int count = 0;
        while (true) {
            System.out.print("Enter Admission Number: ");
            int id = input.nextInt();
            boolean exists = false;
            for (int i = 0; i < count; i++) {
                if (ids[i] == id) {
                    exists = true;
                    break;
                }
            }
            if (exists) {
                System.out.println("Duplicate Admission number...");
                System.out.print("Do you want to enter another admission number? (y/n): ");
                char choice = input.next().charAt(0);
                if (choice == 'n' || choice == 'N') {
                    break;
                }
            } else {
                input.nextLine();
                System.out.print("Enter Student Name: ");
                String name = input.nextLine();
                ids[count] = id;
                names[count] = name;
                count++;
                System.out.print("Do you want to add another student? (y/n): ");
                char choice = input.next().charAt(0);
                if (choice == 'n' || choice == 'N') {
                    break;
                }
            }
        }
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter Admission Number: 2342
Enter Student Name: fd
Do you want to add another student? (y/n): n

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter Admission Number: 2342342
Enter Student Name: sf
Do you want to add another student? (y/n): y
Enter Admission Number: 232434
Enter Student Name: dfdddd
Do you want to add another student? (y/n): n

-----
(program exited with code: 0)
```

Q40.

```
import java.util.*;
class Example40 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int[] clothingIDs = new int[1000];
        int count = 0;
        while (true) {
            System.out.print("Enter Clothing ID (-1 to stop): ");
            int id = input.nextInt();

            if (id == -1) {
                break;
            }
            boolean isDuplicate = false;
            for (int i = 0; i < count; i++) {
                if (clothingIDs[i] == id) {
                    isDuplicate = true;
                    break;
                }
            }
            if (isDuplicate) {
                System.out.println("Duplicate element! ID already exists.");
            } else {
                clothingIDs[count] = id;
                count++;
            }
        }
        int[] finalArray = new int[count];
        for (int i = 0; i < count; i++) {
            finalArray[i] = clothingIDs[i];
        }
        System.out.println("Selected Clothing IDs: " + Arrays.toString(finalArray));
    }
}
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter Clothing ID (-1 to stop): 22342
Enter Clothing ID (-1 to stop): 234322
Enter Clothing ID (-1 to stop): 23233
Enter Clothing ID (-1 to stop): 2322
Enter Clothing ID (-1 to stop): 23224
Enter Clothing ID (-1 to stop): 2322
Duplicate element! ID already exists.
Enter Clothing ID (-1 to stop): 23243
Enter Clothing ID (-1 to stop): 2323
Enter Clothing ID (-1 to stop): -1
Selected Clothing IDs: [22342, 234322, 23233, 2322, 23224, 23243, 2323]
```

(program exited with code: 0)

Press any key to continue . . . |